ecommerce-microservice

Run the stack locally, verify it end-to-end, and exercise a real Stripe test-mode payment — plus the new Kafka UI.

Contents

Prerequisites

Java	17+ (JDK)
Maven	any recent 3.x (`mvn -v`)
Docker	Docker Desktop / Docker Engine with Compose v2 (`docker compose version`)
Node.js	only needed if you also want to run the frontend (`cd frontend && npm run dev`)
Stripe CLI	only needed for the *full* real payment test — docs.stripe.com/stripe-cli, then `stripe login`

> Only infrastructure that's painful to run locally is dockerized: Zipkin, Kafka, Kafka UI, and the 4 per-service Postgres databases. The 8 Spring Boot services run as plain local JVMs via `mvn spring-boot:run` — this makes debugging, breakpoints, and hot-reload straightforward.

1 Start infrastructure (Docker)

From the repo root:

```
cp .env.example .env   # fill in JWT_SECRET, STRIPE_SECRET_KEY,
STRIPE_WEBHOOK_SECRET_KEY, POSTGRES_PASSWORD
docker compose up -d
```

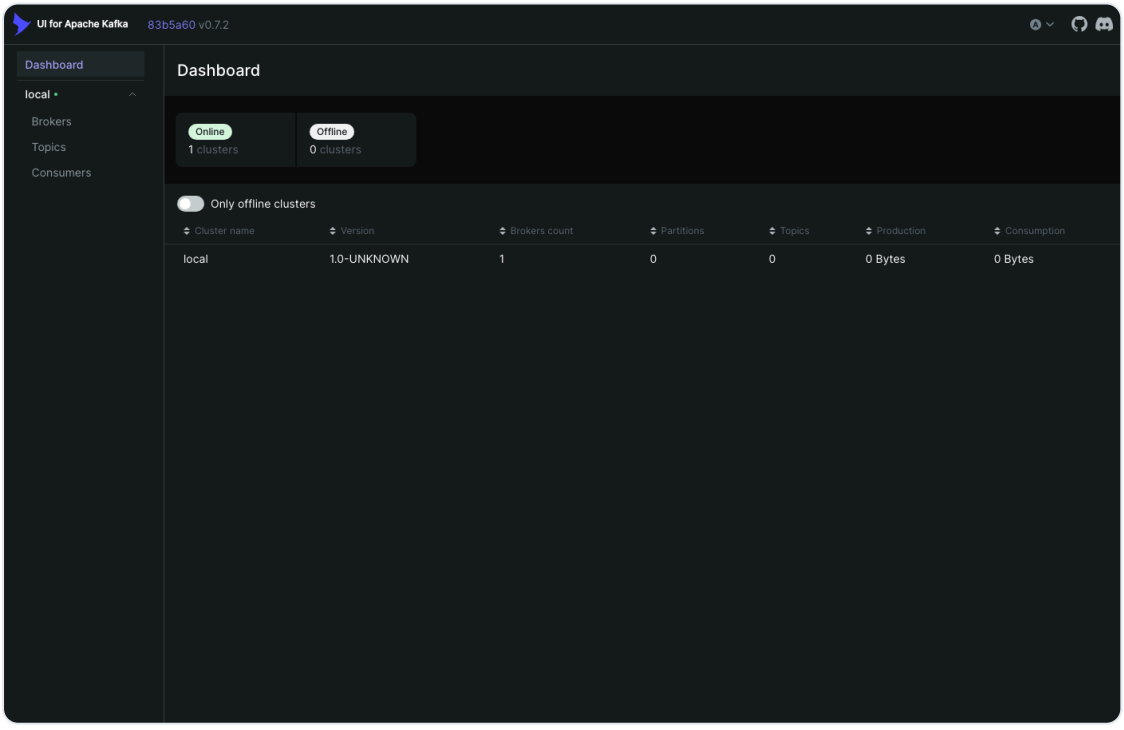
This starts 7 containers:

Container	Purpose	Port
`zipkin`	distributed tracing UI	9411
`kafka`	event broker (KRaft mode, no Zookeeper)	9092 (host), 29092 (internal)
`kafka-ui`	web UI for browsing topics/messages	8090
`user-db`	Postgres for user-service	5433
`catalog-db`	Postgres for catalog-service	5434
`cart-db`	Postgres for cart-service	5435
`order-db`	Postgres for order-service	5436

> **Why Kafka has two listeners:** Kafka tells clients where to reconnect via its "advertised" address. Host-run services (`order-service` , `payment-service`) need to reach it at `localhost:9092` ; other containers (`kafka-ui`) need to reach it at `kafka:29092` over the compose network. One listener can't correctly serve both audiences, so `docker-compose.yml` defines `PLAINTEXT` (host, :9092) and `INTERNAL` (containers, :29092) side by side.

2 Kafka UI

Once infra is up, open `http://localhost:8090` . You'll see the `local` cluster online, with 0 topics until services start creating/using them.



Once `order-service` and `payment-service` are running (next section) and you've triggered a checkout, use **Topics** → **payment-events** → **Messages** to watch the actual `PaymentEvent` JSON

(`{"orderId":...,"status":"PAID"}`) that `payment-service` published and `order-service` consumed — this is the easiest way to see the async hop in the system happen live.

3 Start the 8 services locally

Order matters: `eureka-server` → `config-server` → everything else. Each service's `application.yml` already defaults to `localhost` for every dependency, matching the ports Docker Compose exposes.

```
set -a; source .env; set +a   # exports JWT_SECRET, STRIPE_SECRET_KEY, etc.

mvn -f services/eureka-server/pom.xml spring-boot:run &
mvn -f services/config-server/pom.xml spring-boot:run &
# wait for both to print "Started ...Application", then:
mvn -f services/user-service/pom.xml spring-boot:run &
mvn -f services/catalog-service/pom.xml spring-boot:run &
mvn -f services/cart-service/pom.xml spring-boot:run &
mvn -f services/order-service/pom.xml spring-boot:run &
mvn -f services/payment-service/pom.xml spring-boot:run &
mvn -f services/api-gateway/pom.xml spring-boot:run &
```

Useful URLs once everything is up:

App / API gateway	`http://localhost:8080`
Eureka dashboard	`http://localhost:8761`
Config server	`http://localhost:8888/{service-name}/default`
Zipkin	`http://localhost:9411`
Kafka UI	`http://localhost:8090`

> Right after startup, the gateway's internal load-balancer cache can lag a few seconds behind Eureka's actual registrations — a request in that window may return `503` . If that happens, just retry a couple of seconds later; no restart needed.

4 Smoke-test the app

```
# Confirm the gateway can reach catalog-service
curl http://localhost:8080/api/products

# Confirm every service registered with Eureka (expect 7 entries)
curl -s http://localhost:8761/eureka/apps -H "Accept: application/json" \
  | python3 -c "import json,sys; d=json.load(sys.stdin); print([a['name'] for a in
d['applications']['application']])"
```

A Postman collection covering every endpoint (auth, catalog, cart, checkout, payments) is at `docs/ecommerce-microservice.postman_collection.json` — import it into Postman for interactive

testing, or run it headlessly with `newman run docs/ecommerce-microservice.postman_collection.json --env-var baseUrl=http://localhost:8080`.

5 Quick payment sanity check (no real Stripe needed)

With the dummy keys from `.env.example`, you can verify the checkout flow is wired correctly end-to-end up to the Stripe API boundary, without any Stripe account:

```
TOKEN=$(curl -s -X POST http://localhost:8080/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"you@example.com","password":"..."}' | python3 -c "import
json,sys;print(json.load(sys.stdin)['token'])")

CART_ID=$(curl -s -X POST http://localhost:8080/api/carts | python3 -c "import
json,sys;print(json.load(sys.stdin)['id'])")
curl -s -X POST http://localhost:8080/api/carts/$CART_ID/items -H "Content-Type:
application/json" -d '{"productId":1}'

curl -s -X POST http://localhost:8080/api/checkout \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d "{\"cartId\":\"$CART_ID\"}"
```

With a dummy `STRIPE_SECRET_KEY`, this correctly fails when payment-service calls the real Stripe API — order-service catches that and rolls back the order it just created. That failure *is* the expected, correct behavior in this mode; it proves the Feign call chain (gateway → order-service → payment-service) and the rollback-on-failure logic both work.

6 Full real payment test (Stripe CLI, test mode)

This exercises the entire real path: a genuine Stripe Checkout Session, a real (test-mode) card payment, a signed webhook delivered back to `payment-service`, a Kafka event, and `order-service` updating the order to `PAID`.

6.1 — Log in to the Stripe CLI (test mode)

```
stripe login
stripe config --list   # note the "test_mode_api_key" (sk_test_...) for the next step
```

6.2 — Start the webhook forwarder

```
stripe listen --forward-to http://localhost:8080/api/checkout/webhook
```

Copy the `whsec_...` secret it prints — you'll need it below. Leave this command running in its own terminal.

> `STRIPE_WEBHOOK_SECRET_KEY` is *ephemeral*: `stripe listen` generates a new `whsec_...` value every time you start it (unless you set up a fixed webhook endpoint in the Stripe dashboard instead).

> Whichever way you configure it below, you'll need to update this one value each time you restart `stripe listen`.

6.3 — Put the real keys in `.env`

Edit `.env` directly and replace the two dummy values:

```
STRIPE_SECRET_KEY=sk_test_...        # from `stripe config --list`
STRIPE_WEBHOOK_SECRET_KEY=whsec_...  # from `stripe listen` above
```

Since `.env` is gitignored and every service already sources it via `set -a; source .env; set +a` before starting, this is all you need — restart (or start) `payment-service` normally and it picks up the real keys with no extra flags:

```
mvn -f services/payment-service/pom.xml spring-boot:run
```

Prefer not to touch `.env` for a one-off test? You can override just for that shell instead — but the order matters:

```
# kill any already-running payment-service first, then:
set -a; source .env; set +a
export STRIPE_SECRET_KEY=sk_test_...        # override AFTER sourcing .env, not before
export STRIPE_WEBHOOK_SECRET_KEY=whsec_...  # same here
mvn -f services/payment-service/pom.xml spring-boot:run
```

If you export *before* sourcing `.env`, the dummy values in `.env` silently clobber your overrides when it's sourced, and every webhook will fail signature verification with a generic 500 — exporting after sourcing avoids this entirely, or just skip the whole issue by editing `.env` as in 6.3.

6.4 — Check out

Register/login, build a cart, and hit checkout exactly as in the quick check above. This time the response contains a real Stripe-hosted `checkoutUrl` (starts with `https://checkout.stripe.com/...`) and the order actually persists.

6.5 — Pay with a Stripe test card

> **Before you pay, check `WEBSITE_URL` in `.env`.** Stripe redirects the browser to `WEBSITE_URL/checkout-success?orderId=...` after payment. `.env.example` defaults this to `http://localhost:8080` (the gateway) — but the gateway has nothing to serve at that path locally, since there's no Docker build bundling the frontend into it anymore (see "Known limitations" in the README). Hitting it gives a Whitelabel `404`, not a real error — the payment itself still succeeded. Run the frontend standalone (`cd frontend && npm run dev`, defaults to `:5173`, already proxies `/api` to the gateway — see `frontend/vite.config.ts`) and set `WEBSITE_URL=http://localhost:5173` in `.env` before starting `payment-service`, so the redirect actually lands on a page that exists.

Open the `checkoutUrl` in a browser and pay with any Stripe test card, e.g.:

Card number	Expiry	CVC	Result

`4242 4242 4242 4242`	any future date	any 3 digits	✓ succeeds

On success you're redirected to `{WEBSITE_URL}/checkout-success?orderId=...` (e.g. `http://localhost:5173/checkout-success?orderId=42` with the fix above).

6.6 — Verify the webhook and the resulting order status

```
# The stripe listen terminal should show:
#   --> payment_intent.succeeded [...]
#  <--  [200] POST http://localhost:8080/api/checkout/webhook [...]

curl -s http://localhost:8080/api/orders/<orderId> -H "Authorization: Bearer $TOKEN"
# {"id":...,"status":"PAID",...}
```

You can also watch this land in real time in **Kafka UI** → **Topics** → **payment-events** → **Messages** (see section 2).

7 Tearing everything down

```
# stop the 8 local services (Ctrl+C each, or):
pkill -f "spring-boot:run"

# stop stripe listen if you ran the full payment test:
pkill -f "stripe listen"

# stop infra:
docker compose down
```

8. Troubleshooting

Symptom	Likely cause / fix
Gateway returns 503 right after startup	Load-balancer cache hasn't refreshed yet — retry in a few seconds.
Webhook always returns 500 "Error creating a checkout session"	Signature verification failed — usually a stale/wrong `STRIPE_WEBHOOK_SECRET_KEY`. Re-check the value from the currently-running `stripe listen` session (it changes each time you restart `stripe listen`), and make sure you exported it *after* sourcing `.env`.
Checkout succeeds but order stays PENDING	Give the Kafka consumer a couple of seconds, or check `payment-events` in Kafka UI to confirm the event was actually published.
Kafka UI shows the cluster offline	Confirm `kafka-ui`'s `KAFKA_CLUSTERS_0_BOOTSTRAPSERVERS` is `kafka:29092` (the INTERNAL listener), not `localhost:9092` — the latter only resolves for host processes, not other containers.

"No servers available for service: X" in gateway logs	That service hasn't registered with Eureka yet, or crashed on startup — check its own log and `http://localhost:8761`.
Whitelabel Error Page / 404 after paying, at `/checkout-success?orderId=...`	The payment itself succeeded — this is just `WEBSITE_URL` pointing at the gateway instead of the running frontend. See the callout in 6.5: run the frontend (`npm run dev`) and set `WEBSITE_URL=http://localhost:5173`.

Generated for the ecommerce-microservice project. See README.md for architecture details, and docs/system-design.drawio / docs/ecommerce-microservice.postman_collection.json for the diagram and API collection.